The class number of the CM field of NIST P-224 (and P-256)

2026-10-22

Status

Estimate plus exact genus theory; no exact class number. Protocol: PROTOCOL.md, committed before the computation. Code: `src/cryptanalysis/class_number_estimate.rs` and `examples/class_number_estimate.rs`. Records: run/results.json, run/convergence.svg.

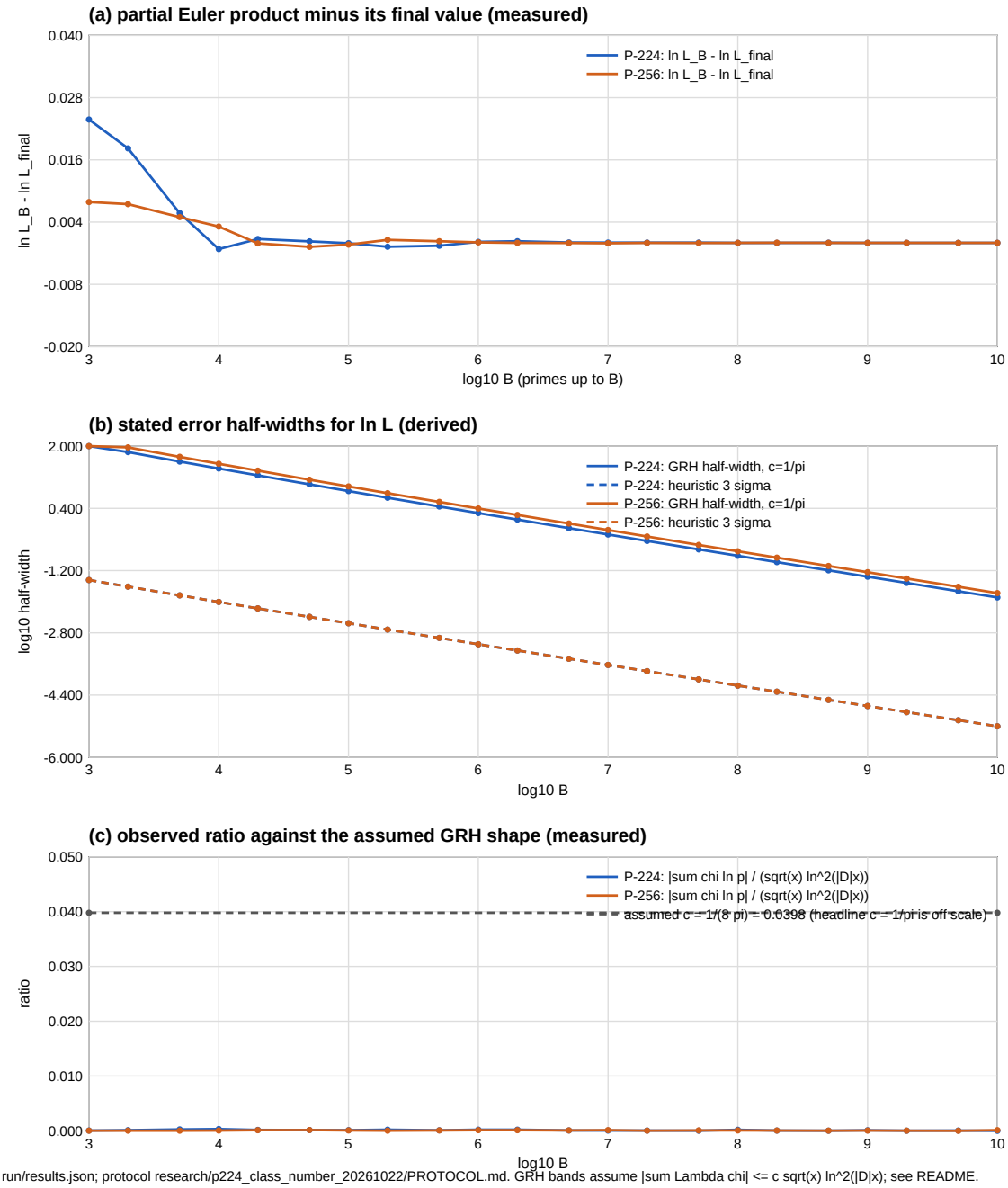
This closes the item "class number of `D_K`: open" in `research/p224_hypothesis_ranking_20261008/README.md` as far as is honest: the size of `h(D_K)` is now known to a stated interval under stated assumptions, and its 2-part is known exactly. The exact value is **not** known and is not computable here (section "Can `h` be computed exactly?").

This is not an ECDLP cost measurement. It has no rho ratio, no phase is priced, no scoreboard row is owed, and it is not an improvement of the index-calculus baseline (`AGENTS.md` section 7a and the success criterion). A bound on the size of a class group says nothing, by itself, about the hardness of the curve's discrete logarithm.

Evidence labels: **EXACT** (a theorem plus exact integer arithmetic), **MEASURED** (computed by this run), **GRH-COND.** (valid if GRH holds, and where stated if a named constant holds), **HEURISTIC** (random model), **DERIVED** (arithmetic on the above).

Result

Euler-product convergence for the CM discriminants of P-224 and P-256



run/results.json; protocol research/p224_class_number_20261022/PROTOCOL.md. GRH bands assume |sum Lambda chi| <= c sqrt(x) ln^2(|D|x); see README.

Euler-product convergence, the stated error half-widths, and the observed ratio against the assumed GRH shape. Source: `run/results.json`, example `class_number_estimate`, B = 10^10, 4 threads.

	P-224	P-256
D_K	-3·29·79·7523·40927·11549194661·L_{45} (L_{45} = 38842507490385260348140872723572577484402299)	-3·5·456597257999·1428624589419343516204097·46523541035814968339936406074986559003387
conductor of Z[π]	3	1
$\log_2 \lvert D_K \rvert$	222.494	257.975
D_K mod 8	5 (so χ(2) = -1)	5 (so χ(2) = -1)
prime discriminants t	**7**	**5**
2-rank of Cl(D_K)	**6** (EXACT)	**4** (EXACT)
Rédei 4-rank	**0** (EXACT)	**0** (EXACT)
2-part of Cl	(Z/2)^6, so h = 64·m, m odd	(Z/2)^4, so h = 16·m, m odd
L(1,χ), Euler product to B = 10^10	0.358762 (MEASURED)	0.722539 (MEASURED)
h central value (√|D| L/π)	**3.5185 × 10^32 = 2^108.1167**	**1.5517 × 10^38 = 2^126.8672**
HEURISTIC 3σ interval (relative ±6.3×10^{-6} in ln L)	[3.5185, 3.5185] × 10^32	[1.5517, 1.5518] × 10^38
GRH-COND. interval, assumed c = 1/(8π)	[3.5128, 3.5242] × 10^32 (2^108.1143 to 2^108.1190)	[1.5485, 1.5550] × 10^38
GRH-COND. interval, assumed c = 1/π (headline)	**[3.4735, 3.5641] × 10^32 (2^108.098 to 2^108.135)**	**[1.5261, 1.5778] × 10^38 (2^126.843 to 2^126.891)**
sweep wall time (4 threads, shared host)	93 s	93 s

Reading the table:

- The headline interval for h(D_K) of P-224 is **[3.47, 3.56] × 10^32, i.e. 2^108.10 to 2^108.14**, **conditional on GRH and on the assumed explicit constant c = 1/π** below. It sits inside the earlier "2^106 to 2^111" range and above the "GRH asymptotic lower bound of about 2^106". The earlier text's 2^109.6 · L(1,χ) is now 2^109.6 × 0.3588 = 2^108.1. L(1,χ) is small because 2 is inert (D ≡ 5 mod 8) and 3 is ramified, which removes the factors (1+1/2)^-1 and (1-1/3)^-1.
- The D ≡ 1 mod 4, odd, squarefree shape and the identity $t^2 - 4p = -f^2 \cdot \prod \text{factors}$ were re-checked by the program before the sweep (a mismatch would stop it). All factors pass a 12-base strong probable-prime test here. For P-224 the earlier report records them as proved (cm_discriminant.json); for P-256 they are the SafeCurves factorisation, "probable" in published_factorizations.json. Genus theory needs the factors to be prime.
- **Nothing about the interval is PROVED.** What is proved: the 2-rank, the Rédei 4-rank and so the 2-adic valuation $v_2(h)$.

Method

1. Euler product (MEASURED)

χ(p) = (D_K/p) for every prime p ≤ B = 10^10 (about 4.55 × 10^8 primes), by a segmented sieve and a Jacobi symbol of D_K mod p (limbs of |D_K| in base 2^29, u64 reduction). The result is deterministic: segments are merged in order with Kahan summation, and a unit test checks that 1 and 3 threads give bit-identical sums. ln L_B = Σ_{p≤B} −ln(1 − χ(p)/p), then h = √|D_K| · L / π (w = 2).

Convergence (run/results.json, checkpoints; panel (a) of the figure): ln L_B for P-224 is −1.00132 at B = 10^3, −1.02631 at 10^4, −1.02510 at 10^6, and −1.0250974 at 10^10; the last decade moves it by less than 5 × 10^-6. About half the primes split and half are inert at every checkpoint (P-224 at 10^10: 227,526,100 split, 227,526,406 inert, 5 dividing D: 3, 29, 79, 7523, 40927; the two largest prime factors exceed B).

2. Error statements

- **HEURISTIC.** If the signs χ(p), p > B, behave as independent fair signs, the tail of ln L has standard deviation (Σ_{p>B} p^-2)^{1/2} ≈ 1/√(B ln B): 2.1 × 10^-6 at B = 10^10. This is a model, not a theorem. Calibration (below) supports it for discriminants of size 10^7 at small B.
- **GRH-COND., with a named assumed constant.** Assume |Σ_{n≤x} Λ(n) χ(n)| ≤ c √x ln²(|D| x) for x ≥ B. The constant c = 1/(8π) is the Schoenfeld-type shape I recall for this kind of bound; I did not re-derive it or look it up here, so the headline uses c = 1/π, eight times larger, as a safety factor. Then, with F(t) = c√t ln²(|D|t) + 1.2√t (the second term covers prime powers, ψ(t) − θ(t) < 1.2√t for t ≥ 1000), partial summation gives |ln L − ln L_B| ≤ F(B)/(B ln B) + ∫_B^∞ F(t)(ln t + 1)/(t² ln² t) dt + 1/B, where the last term bounds the $u^2/2$ Euler-factor remainder. The integral is evaluated numerically (Simpson in ln t, 24,000 steps to t = B e^120). At B = 10^10 the half-width of ln L is 0.0129 (c = 1/π) and 0.0016 (c = 1/(8π)) for P-224, 0.0167 and 0.0021 for P-256. The bound decays like ln²|D| / (√B ln B); panel (b) of the figure shows both half-widths against B.
- **Observed against the assumed shape (MEASURED).** The measured maximum of |Σ_{p≤x} χ(p) ln p| / (√x ln²(|D|x)) over the 21 checkpoints is 3.1 × 10^-4 for P-224 and 1.3 × 10^-4 for P-256, about 130 times below 1/(8π) = 0.0398. This is a check on the plausibility of the assumed form, not a verification of it for all x.

3. Controls (pre-registered; run/results.json, controls)

400 consecutive fundamental discriminants below −10^7 (D from −10,000,003 to −10,001,323), exact h by counting reduced forms:

B	within 3σ (heuristic)	GRH band (c = 1/π) covers	sd of z = (ln h − ln h_B)/σ	max |z|
10^3	99.75 % (399/400)	100 %	0.85	3.11
10^4	100 %	100 %	0.84	2.81
10^5	100 %	100 %	0.93	2.31

The pre-registered criteria (GRH band covers every D; 3σ covers at least 99 %) hold. The heuristic σ is not underestimated on these small discriminants (sd of z is below 1). That is a calibration at |D| ~ 10^7, B ≤ 10^5; it does not establish the model at |D| ~ 2^222, B = 10^10, where B is

astronomically below √|D|. Further controls, all unit tests in the module: exact class numbers of 23 tabulated discriminants (−3, −4, −7, −8, −11, −15, −20, −23, −31, −39, −47, −71, −79, −84, −87, −95, −103, −127, −151, −163, −199, −239, −427: h = 1,1,1,1,1,2,2,3,3,4,5,7,5,4,6,8,5,5,7,1,9,15,2); the theta series agrees with exact `h` to 10^-8 for every fundamental `D > −3000`; the Euler product agrees with exact `h` within 1.5 % at `B = 10^6` for four `D` near 10^4 to 10^6; genus theory gives 0 violations on every fundamental `D > −3000` and on the 400 control discriminants.

4. Cohen-type series (cross-check only)

`L(1,χ) = Σ χ(n) e^{−πn²/|D|}/n + (π/√|D|) Σ χ(n) erfc(n √(π/|D|))`, from the functional equation of the odd quadratic character (root number 1). The terms fall below 10^-17 once `n > √(12.4|D|)`, so it needs about √|D| terms. It is exact to double precision for small `D` and is what the unit test compares against exact `h`. At `|D_K|` it needs about `√(12.4·|D_K|) = 2^113.1` terms: **not runnable**. The Euler product is the only analytic method used for `D_K`.

5. Genus theory and the Rédei 4-rank (EXACT)

For a negative fundamental discriminant with `t` prime discriminants the class group has 2-rank `t − 1` (genus theory). `D_K` of P-224 is odd with 7 prime factors, so `Cl[2] ≅ (Z/2)^6` and **`2^6 | h`, `h` even**; of P-256, 5 factors, `(Z/2)^4`, `2^4 | h`. The 4-rank is `t − 1 − rank_F2(R)` for the Rédei matrix `R` (off-diagonal `r_ij = 0` iff `(d_j / p_i) = +1` for the prime discriminants `d_j`, diagonal = row sum). The rank is 6 (P-224) and 4 (P-256), so the 4-rank is **0 in both**: the 2-Sylow subgroup is elementary abelian, `v₂(h) = t − 1` exactly, and `h / 2^(t-1)` is odd. The method is checked on every fundamental `D > −3000` against `v₂` of the exact class number (4-rank 0 implies `v₂ = t − 1`; positive 4-rank implies `v₂ ≥ t`; the control is not vacuous, more than 20 discriminants have positive 4-rank).

DERIVED consequence: the odd cofactor `h/64` is about 5.50 × 10^30 (P-224) and `h/16` about 9.70 × 10^36 (P-256), each with the relative uncertainty of the `h` interval. The 8-rank and higher 2-power structure are moot because the 2-Sylow subgroup is already `(Z/2)^{t-1}`. Nothing is said here about the odd part's structure, in particular whether `Cl(D_K)` is cyclic away from 2.

Can `h` be computed exactly?

Not here, and not unconditionally anywhere I can point to. Nothing in this section was run; the operation counts are scalings from the literature, with the `o(1)` terms dropped, not measurements, and I did not look the published constants up while writing it.

- **Analytic formula, exact.** Cohen's series needs `2^113` terms (above). Reaching `h` exactly from `L(1,χ)` needs `L` to relative precision about `1/h = 2^-108`, so the truncated Euler product, whose error is a conditional `~10^-2` and a heuristic `~10^-6`, is about 30 orders of magnitude short.
- **Unconditional.** A proof that a candidate value equals `h` without GRH needs either every prime ideal up to the Minkowski bound (`√(|D|/3) ≈ 2^110`) or an exponential-time algorithm. Shanks-type baby-step giant-step in the class group costs about `√W` group compositions for an enclosure of width `W`. With no enclosure `W ~ h ~ 2^108` and `√W ~ 2^54`, i.e. order `|D|^{1/4}`; at 1 to 5 µs per 222-bit composition (an assumed figure) that is about 1.8 × 10^16 compositions, i.e. 600 to 3,000 core-years. Even a fully trusted enclosure does not make it small: the GRH-conditional `c = 1/π` interval has `W ≈ 2^102.8` and `√W ≈ 2^51.4`; the heuristic 3σ interval has `W ≈ 2^91.8` and `√W ≈ 2^45.9`, still about 2 to 11 core-years of compositions with a baby-step table of `2^46` entries that does not fit in 15 GB (a memoryless kangaroo variant avoids the table but not the time, and a conditional enclosure makes the result conditional).
- **Subexponential (Hafner–McCurley, Buchmann, Jacobson).** The cost scales as `L_|D|[1/2, c] = exp(c √(ln|D| ln ln|D|))`: here `ln|D| = 154.2`, `√(ln|D| ln ln|D|) = 27.9`, so `c = 1` gives `e^27.9 ≈ 2^40` and `c = √2` gives `e^39.4 ≈ 2^57` (P-256: `e^30.5 ≈ 2^44` and `e^43.1 ≈ 2^62`). As I recall, published implementations reach discriminants of about 100 digits, and `|D_K|` has 67 digits, so the class of algorithm is in range for this size on a multi-core machine; I make no claim about how long it would take on this host. **It is not available here:** this host has no PARI/GP, Magma or Sage, and `AGENTS.md` requires the computation to be Rust. Writing a relation-collecting class-group engine (factor base up to the GRH bound of order `12 ln²|D| ≈ 2.9 × 10^5`, sparse linear algebra over Z, Hermite or Smith normal form of the relation lattice) is a project of its own, and I did not start it.
- **What the result would be.** The relation-lattice determinant is a multiple of `h`; it equals `h` if it lies within a factor 2 of an independent estimate of `h`. The estimate above (error ≲ 2.6 % under GRH) is already far tighter than that step needs. But that makes the final `h` **GRH-conditional**: the factor-base bound that makes the relations generate `Cl(D_K)` is itself a GRH statement. I know of no way to turn it into an unconditional proof at 222 bits.

So the honest summary is: an interval (conditional) and the exact 2-part are what can be had here; a GRH-conditional exact `h` is a feasible but unbuilt project; an unconditional exact `h` is out of reach of any known method at this size.

Limits

- The GRH-conditional intervals assume an explicit constant that I did not derive (`c = 1/π`, eight times the shape quoted from memory). If the true explicit bound is weaker, the interval widens; the measured shape ratio (130 times below `1/(8π)`) makes that unlikely but is not proof.
- The heuristic interval is a random-sign model; its calibration is on small `D` and small `B`.
- The sweep used `B = 10^10` only (protocol). Extending to `B = 10^11` would shrink the conditional half-width by about √10, at about 15 minutes of 4-core time per curve; not done.
- Primality of the factors is the 12-base strong test here; the Rédei and genus statements are exact given that the listed factors are prime.
- Wall times are from a shared host and are not a benchmark. No timing is claimed as a result, and nothing goes to `benchmarks/`.

Reproduce

```
cargo test --release --lib cryptanalysis::class_number_estimate
cargo run --release --example class_number_estimate -- \
    --curve both --bound 10000000000 --threads 4 --out /tmp/cn-check
# compare /tmp/cn-check/results.json with run/results.json (identical except
# sweep_wall_seconds)
```

SHA-256 of the committed outputs:

```
61016700c9aeef3478c3d9ca11425a653a7dd09b177ca7874648aefefe1e19d5  run/results.json
74d5a0e907eb6b479081f190ee24e9fbdf68d2cdcf87795dc95941247020158b  run/convergence.svg
```